

Direct AMY LPM Transport and Full-Angle Screened Thermalisation

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Archive reconstruction from the surviving project ledger. Historical sandbox binaries were ephemeral;
see RECOVERY_PROVENANCE.md.

50. Direct AMY upgrade

The v1.4 calculation directly solved the isotropic AMY transverse LPM equation for the QCD splitting channels and replaced angle-averaged elastic scattering by deterministic full-angle screened quadrature. At the reheating benchmark

$$\frac{M_D}{T} = 0.01, \quad \alpha_s = 0.0393544, \quad y_D = 0.30, \quad \frac{p}{T} = 3,$$

the dimensionless rates are

$$\frac{\Gamma_{g \rightarrow gg}}{T} = 0.261356,$$

$$\frac{\Gamma_{q \rightarrow gq}}{T} = 0.204516, \quad \frac{\Gamma_{D \rightarrow gD}}{T} = 0.201417,$$

$$\frac{\Gamma_{g,\text{elastic}}}{T} = 0.106177, \quad \frac{\Gamma_{q,\text{elastic}}}{T} = 0.0119151,$$

and

$$\boxed{\frac{\Gamma_{H \leftrightarrow qD}}{T} = 3.18544 \times 10^{-4}.}$$

The QCD rates are fast. The slow mode is the scalar-Yukawa conversion $H \leftrightarrow qD$. At

$$T_0 = 1.002 \times 10^8 \text{ GeV},$$

the kinetic rate is

$$\Gamma_{\text{kin}} = 3.1918 \times 10^4 \text{ GeV}.$$

Compared with

$$\Gamma_R = 1.47850065 \times 10^{-2} \text{ GeV},$$

one obtains

$$\boxed{\frac{\Gamma_{\text{kin}}}{\Gamma_R} = 2.1588 \times 10^6.}$$

The slow portal conversion remains more than six orders of magnitude faster than reheaton decay.

Figure 1: The slow portal conversion remains more than six orders of magnitude faster than reheaton decay.

Even a conservative factor-of-two normalisation uncertainty leaves 1.0794×10^6 . The correction to the desired reheating branch is negligible.

51. What v1.4 did and did not solve

The direct AMY calculation substantially closes the QCD kinetic problem. It does not yet contain the complete electroweak/Yukawa LPM structure of $H \leftrightarrow qD$. The outstanding calculation requires:

1. the complete helicity and Higgs-doublet source structure;
2. simultaneous $SU(3)_c$, $SU(2)_L$, and $U(1)_Y$ soft collision kernels;
3. thermal widths and asymptotic masses;
4. exact matching to the scalar retarded self-energy;
5. insertion into a reduced gauge-covariant Schwinger-Keldysh model.

Only after that does a full $3+1$ dimensional non-Abelian 2PI/Kadanoff-Baym computation become the rational next expenditure. The QCD sector is already far too fast to be the weak link.